

Fan Xiao

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Education

Ludwig-Maximilians-Universität München, Munich, Germany PhD candidate in Medical Physics <i>Research focus:</i> AI-based Efficient Sequence Modeling, Monte Carlo Simulation, Medical Image Segmentation	2023.09 – Now
Southern Medical University, Guangzhou, China MS in Biomedical Engineering, GPA 3.6/4.0	2020.09 – 2023.06
Wuhan University of Technology, Wuhan, China BS in Communication Engineering	2016.09 – 2020.06

Research Experience

Multi-model study of fast VMAT segment dose calculation First Author - Designed a GPU-accelerated processing pipeline using CuPy to perform Ray-Casting for segment projection and Cubic B-spline resampling for Beam's Eye View transformations - Developed a unified benchmarking framework comparing CNN-Mamba, Transformer, CNN-ConvLSTM, and Unet for volumetric sequence modeling (high-resolution dense regression task) - Developed an optimized inference pipeline integrating CUDA Graphs and Automatic Mixed Precision (AMP), achieving high-throughput inference (<40 ms/segment) with low memory footprint (2.9 GB), validated across multiple GPU generations - Published in <i>Physics in Medicine & Biology</i> (2026), DOI: 10.1088/1361-6560/ae6413	2025.07–2026.04
DoseRAD2026 Challenge dataset First Author - Generate a public benchmark of paired CT and MRI data with beam-level photon and proton Monte Carlo dose distributions for developing and evaluating advanced dose calculation methods - Pre-processing included deformable image registration, air-cavity correction, and resampling - Ground-truth photon (6 MV) and proton dose distributions were computed using open-source Geant4 Monte Carlo simulations, yielding 40,500 photon beams and 81,000 proton beamlets - Submitted in <i>Medical Physics</i> (2026)	2025.07–2026.04
Real-Time MRI-Only Photon Dose Calculation Engine First Author - Proposed a CNN-RNN framework (UNet-LSTM) to approximate computationally expensive Monte Carlo particle transport simulations directly from MRI data - Solved the domain shift challenge (lack of electron density information in MRI) by integrating CT-MRI deformable registration with a segmentation-guided feature mechanism - Achieved 12 ms per photon beam inference latency with a gamma passing rate (2%/2mm) exceeding 99.50% compared to Monte Carlo ground truth - Published in <i>Medical Physics</i> (2025), DOI: 10.1002/mp.70106	2024.10 – 2025.06
Proton Range Verification via Prompt Gamma Emission Prediction First Author - Architected an LSTM model by integrating heterogeneous data streams (anatomical relative stopping power maps and physical dose fields) to predict prompt gamma emissions across wide proton energies	2024.01 – 2024.09

(125–210 MeV).

- Achieved sub-millimeter accuracy with a mean absolute range shift deviation of 0.15 mm against Monte Carlo ground truth.
 - Reduced prediction time to <130 ms per pencil beam, facilitating sub-second feedback for adaptive proton therapy.
 - Published in *Physics in Medicine & Biology* (2024), DOI: [10.1088/1361-6560/ad8e2a](https://doi.org/10.1088/1361-6560/ad8e2a)
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Technical Skills

- Languages: Python, C++ (Geant4 focus), Bash.
 - Deep Learning: Pytorch, RNN/LSTM, Transformer, Mamba, Unet/nnUnet.
 - Tools: Git, Docker, Slurm, Cupy, Plastimatch, Pydicom, RayStation scripting.
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Additional Info and Awards

- Conference experience:
ESTRO 2026 Poster (Stockholm), AAPM 2025 Oral (Washington, DC), ICCR 2024 Oral (Lyon), ECMP 2024 Oral (Munich).
 - Awards:
Chinese National Scholarship (2022) – Top 2%;
First-class Academic Scholarship (2022, 2023) – Top 20%.
 - Languages:
Chinese (Native), English (Professional Working Proficiency).
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